AI Fact-Checking System

Group ID
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Abstract

We present a resource-aware climate fact-checking system for realistic limited-compute settings such as Colab-style runtimes. The pipeline stages evidence processing: sparse fusion preserves broad recall, embedding plus factual-cue reranking compresses the pool, a bounded cross-encoder pass selects submitted evidence, and a TF-IDF logistic classifier predicts the verdict from structured evidence text. The resulting system obtains 0.2021 evidence F-score, 0.5000 claim accuracy, 0.2879 harmonic mean, and 0.4659 classifier macro-F1.

1 Introduction

Climate claims are short, compressed, and evidence-sensitive; a useful fact-checker must retrieve evidence as well as predict a label. Practical systems also face limited compute, making staged retrieval and bounded neural scoring central to the design. In this project, the system is fully self-contained at run time and generates ranked pools during notebook execution. The problem is therefore a three-way trade-off among evidence recall, verdict accuracy, and runtime.

Draft note: I collected and grouped the following papers for teammates to continue the related-work writing.

Existing work can be organized into four areas:

1. **Evidence-Based Fact Checking** FEVER frames fact checking as joint evidence retrieval and verdict prediction (Thorne et al., 2018); CLIMATE-FEVER adds climate-specific domain and retrieval challenges (Diggelmann et al., 2020).

2. **Large-Scale Evidence Retrieval** BM25 is fast and scalable (Robertson et al., 1994); reciprocal rank fusion combines ranked lists without supervision (Cormack et al., 2009). These methods provide efficient lexical recall for candidate generation.

3. **Neural Semantic Reranking** Dense embeddings help when lexical overlap is limited (Karpukhin et al., 2020; Reimers and Gurevych, 2019); cross-encoders model claim–evidence interaction directly (Nogueira and Cho, 2019). We place cross-encoder scoring after sparse prefiltering to concentrate neural computation on promising pairs.

4. **Evidence Aggregation and Factual Cue Fusion** Evidence aggregation studies passage combination before classification (Zhou et al., 2019). Hybrid systems show lexical, semantic, and shallow factual signals can be complementary (Lee et al., 2018; Padia et al., 2018).

Our design follows prior retrieval-and-reranking work and targets limited-compute execution. It first builds a broad unsupervised sparse pool from word, character, structured, and PRF gates. It then compresses this pool with MiniLM similarity plus factual features, applies a bounded cross-encoder after prefiltering, and classifies from rank-aware, cue-aware evidence text. Contributions: (i) a staged pipeline for the performance–runtime trade-off, (ii) multi-gate sparse retrieval, (iii) multi-scale reranking that fuses semantic and shallow factual signals, and (iv) diagnostics over sparse gates, fusion, evidence selection, and classifier sensitivity.

2 Method

2.1 Pipeline Overview

Let $\mathcal{C}$ be claims and $\mathcal{E}$ evidence. The output is a verdict and a small evidence set per claim. Since scoring all $(c, e) \in \mathcal{C} \times \mathcal{E}$ pairs is infeasible, we use a staged pipeline:

$$\mathcal{E} \to \mathcal{A}(c) \to \mathcal{B}(c) \to \mathcal{D}(c),$$

where $\mathcal{A}(c)$ is a broad sparse pool, $\mathcal{B}(c)$ a reranked context pool, and $\mathcal{D}(c)$ the submitted evidence.

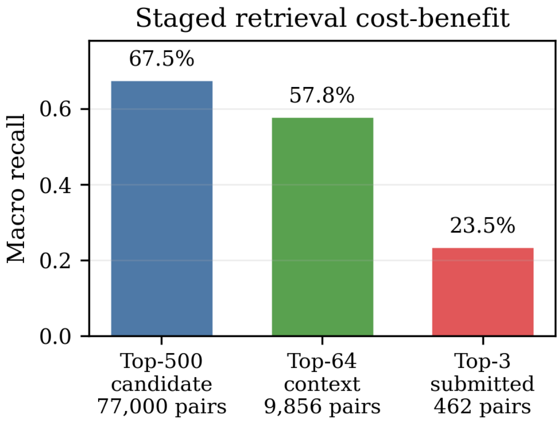

Figure 1: Staged pruning keeps recall while reducing pair count. The candidate and context pools retain most recoverable evidence by using neural scoring on progressively smaller pools; full-corpus neural scoring would require about 186 million development pairs.

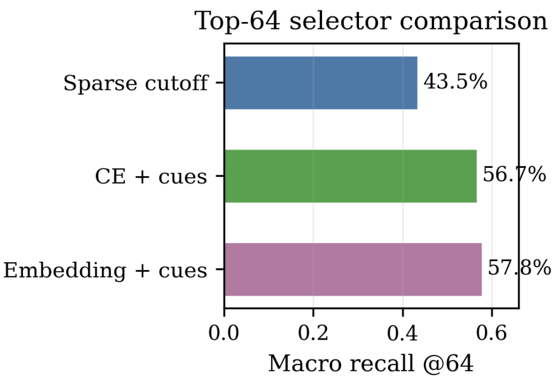

Figure 2: Embedding plus factual cues is the best top-64 selector, reaching 57.8% macro recall above sparse cutoff and cross-encoder-plus-cue variants.

Cheap stages maximize recall; semantic models operate on progressively smaller pools. The four components are multi-gate sparse retrieval, multi-scale reranking, bounded cross-encoder selection, and structured-evidence classification.

2.2 Multi-Gate Sparse Candidate Retrieval

The sparse stage maximizes recall by applying one BM25 scoring operator to multiple feature spaces. The four gates differ in analyzer $a_g(\cdot)$ and vocabulary V_g. Let $q_g(c)$ and $x_g(d)$ be the resulting count vectors.

Word gate. Word terms give the direct lexical view: content-word overlap raises scores and IDF weighting emphasizes discriminative claim terms.

Character gate. Character n-grams capture partial overlap, inflection, hyphenation, and spelling variation missed by exact words.

Structured gate. Structured cues keep numbers, years, salient capitalized terms, and negation markers. These cues add factual constraints that lexical views may lose.

Pseudo-relevance feedback gate. PRF expands the word query using high-ranked feedback evidence. If $\mathcal{F}(c)$ is the feedback set,

$$q_{\text{prf}}(c) = q_{\text{word}}(c) + \eta \sum_{d \in \mathcal{F}(c)} w(c,d) x_{\text{word}}(d),$$

where $w(c,d)$ downweights lower-ranked feedback documents. PRF adds related terms absent from the original claim while staying in the sparse word space.

Shared BM25 interaction. After feature construction, each gate applies the same BM25 interaction. For term t in document d,

$$\text{idf}_t = \log\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right),$$

$$\text{BM25}_{d,t} = \text{idf}_t \frac{(k_1+1) f_{d,t}}{f_{d,t} + k_1(1 - b + b|d|/\overline{|d|})}.$$

The inverse-document-frequency factor suppresses common features, so rare claim-specific evidence terms contribute more than corpus-wide background terms. The query side remains a count vector, giving the sparse dot product

$$s_g(c,d) = q_g(c)^\top \text{BM25}_g(d).$$

Thus g chooses the representation, and BM25 weights it. The notebook uses SciPy/NumPy sparse matrices, so document-frequency counts, BM25 reweighting, and query–document dot products run in optimized C/C++ kernels and vectorized array operations.

Rank fusion. Each gate produces a ranked list. The fifth sparse view is their weighted RRF fusion:

$$S_{\text{RRF}}(c,d) = \sum_{g \in G} \frac{\lambda_g}{K_{\text{RRF}} + r_g(c,d)},$$

where missing documents contribute zero. We treat λ_g as a reliability prior: stronger standalone gates receive larger weights, and complementary factual gates retain smaller weights. The structured

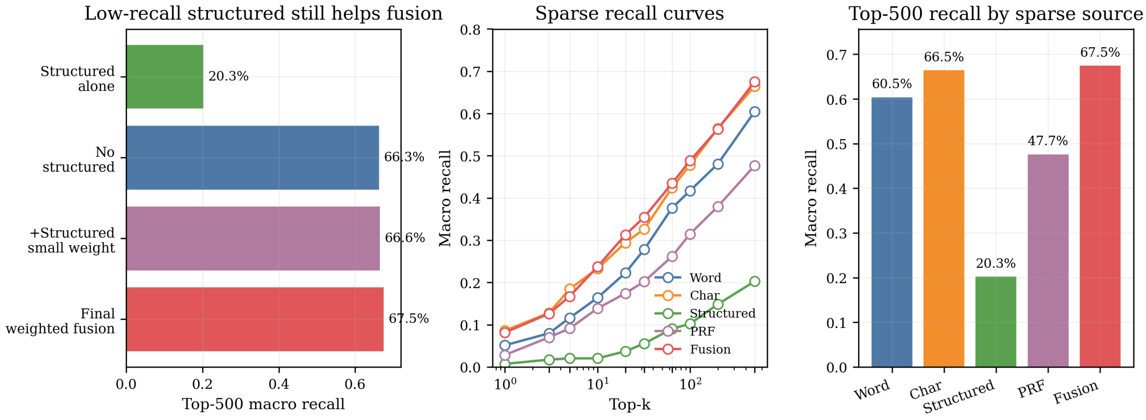

Figure 3: Sparse fusion gives the best top-500 recall and retains structured cues for complementarity. Character matching is the strongest single source; structured cues improve fused recall under controlled ablation, so the final system keeps a small structured weight.

gate contributes complementary recall in fusion, as shown by ablation; RRF also removes the need for cross-gate score calibration.

Figures 1 and 3 justify the candidate stage: fusion improves top-500 recall over any single source, and staging concentrates neural scoring on high-value candidate pools.

2.3 Multi-Scale Feature Fusion

Reranking fuses dense similarity, sparse source rank, word overlap, numeric agreement, negation compatibility, and length shape. MiniLM embeddings are mean-pooled and ℓ_2-normalized, so cosine similarity is an inner product:

$$s_{\text{emb}}(c, d) = h(c)^\top h(d).$$

The shallow factual features are summarized as

$$\phi(c, d) = [\phi_{\text{rank}}, \phi_{\text{lex}}, \phi_{\text{num}}, \phi_{\text{neg}}, \phi_{\text{len}}, s_{\text{emb}}].$$

where the components encode source rank, lexical overlap, number match, negation compatibility, length mismatch, and embedding similarity. A lightweight gradient-boosted matcher estimates $p_{\text{feat}}(c, d)$, and the context score is

$$S_{\text{ctx}}(c, d) = \alpha\, \widetilde{s}_{\text{emb}}(c, d) + (1 - \alpha)\, p_{\text{feat}}(c, d),$$

where $\widetilde{s}_{\text{emb}}$ is claim-wise normalized. Embeddings handle semantic relatedness; factual features guard against number, negation, and lexical-grounding errors.

2.4 Bounded Neural Evidence Selection

The submitted evidence stage uses a stricter operating point than the classifier context. Cross-encoders jointly encode claim and evidence, so we

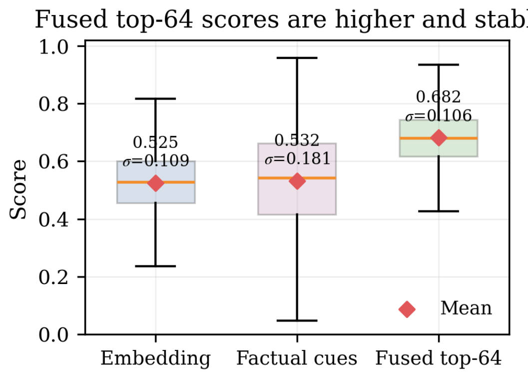

Figure 4: Top-64 fusion raises the retained-candidate score distribution and reduces dispersion. In the Colab diagnostic, fused scores have mean/std = 0.682/0.106, compared with 0.525/0.109 for embeddings and 0.532/0.181 for factual cues.

reserve them for the sparse/embedding-prefiltered pool and normalize their scores within each claim. The final evidence score fuses cross-encoder confidence, embedding confidence, embedding rank, and sparse source rank:

$$S_{\text{sub}}(c, d) = \beta_1 \widetilde{s}_{\text{CE}} + \beta_2 \widetilde{s}_{\text{emb}} + \frac{\beta_3}{\tau + r_{\text{emb}}} + \frac{\beta_4}{\tau + r_{\text{src}}}.$$

Coefficients are fixed. Candidates outside the cross-encoder subset remain ranked by embedding and source-rank terms, which keeps neural cost bounded and retains useful sparse candidates.

Figures 2, 4, and 5 justify multi-scale fusion: factual cues act as complementary signals that improve semantic ranking and stabilize the retained candidate pool.

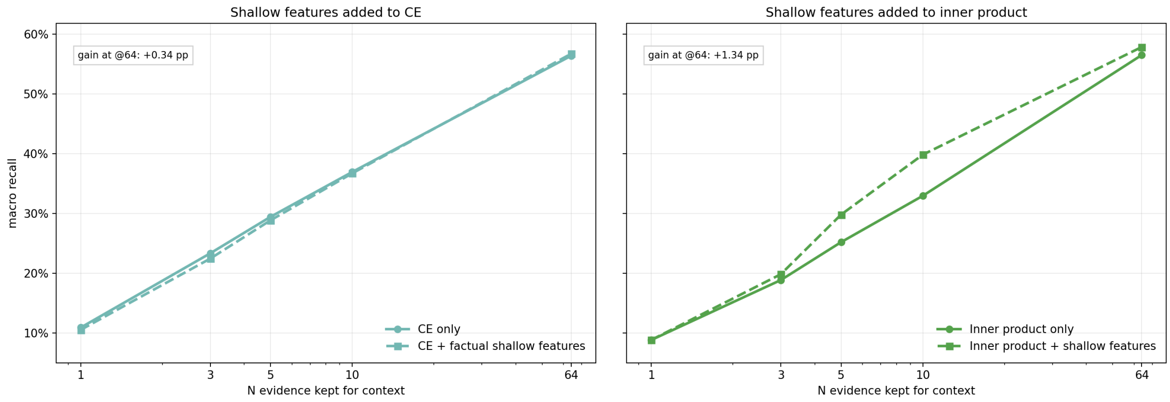

Figure 5: Factual cues preserve or improve semantic ranking. Cue gains are larger for embedding inner product and smaller for the already strong cross-encoder, supporting feature fusion over a purely semantic selector.

2.5 Structured Evidence Classification

The classifier predicts from a bounded ordered evidence context:

$$x_c = [\text{claim}; z_1, z_1, \ldots, z_m, z_m, z_{m+1}, \ldots, z_L],$$

where top evidence is repeated to raise TF-IDF weight. Each evidence line also gets a rank prefix plus word-overlap, numeric-match, and negation-match cue tokens. The logistic model learns weights for these cue tokens from training data. Let v_c be the TF-IDF vector; a one-vs-rest logistic model learns one binary classifier per label:

$$P(y = k \mid c) = \sigma(\theta_k^\top v_c + b_k).$$

The largest decision score gives the label. This keeps classification cheap while preserving retrieval rank and factual signals.

Stage	F-score	Macro recall	Hit-any
Sparse top-500	0.0083	0.6754	0.9026
Context top-64	0.0517	0.5784	0.8636
Submitted top-3	0.2021	0.2355	0.4675

Table 1: Retrieval metrics from the self-generated notebook artifacts.

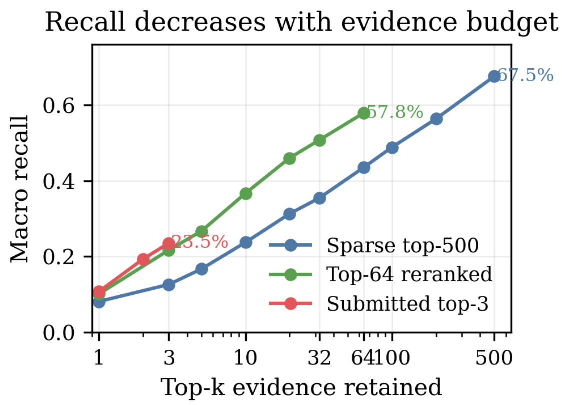

Figure 6: Recall-vs-K curves show the pruning trade-off behind the selected operating points. The sparse pool keeps broad top-500 recall, the reranked pool preserves most recoverable evidence at top-64, and the submitted top-3 is the strict evidence set used for scoring.

3 Results

The final notebook was run in Colab with GPU enabled. Course metrics are evidence F-score F, label accuracy A, and H-score $H = 2FA/(F+A)$. Tables 1–2 report retrieval operating points and reproduced development results.

Figures 6 and 7 show the final retrieval–classification behavior. Remaining classification errors mainly confuse SUPPORTS, REFUTES, and NEI. Label support is uneven, with 18 DISPUTED development examples.

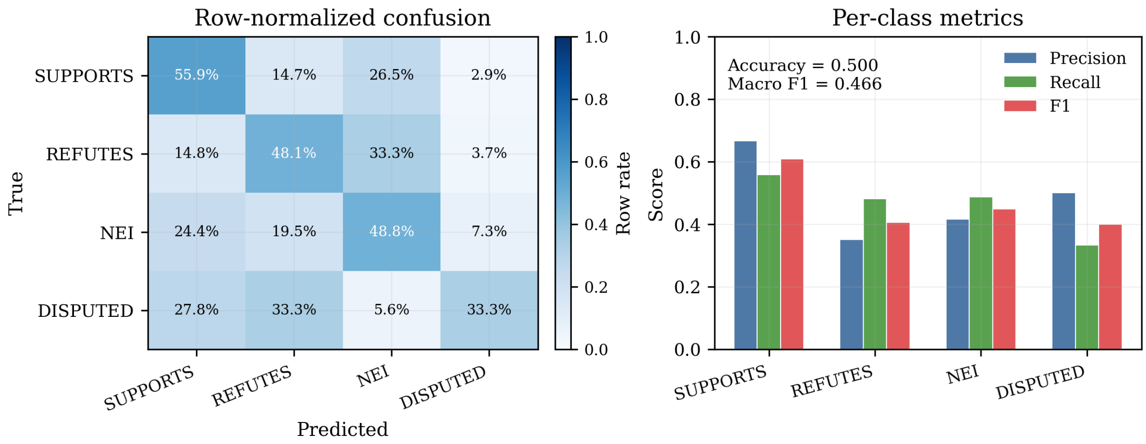

Figure 7: Classifier diagnostics. Row-normalized confusion controls for class support, and per-class precision/recall/F1 summarizes the classification report.

Metric	Value
Evidence Retrieval F-score (F)	0.2021
Claim Classification Accuracy (A)	0.5000
Harmonic Mean of F and A (H)	0.2879
Top-3 Macro Recall	0.2355
Classifier Macro-F1	0.4659

Table 2: Final Colab-reproduced development result. F, A, and $H = 2FA/(F + A)$ are official course metrics; the last two rows summarize retrieval and classification behavior.

Table 3 justifies the classifier input: enhanced context improves over plain context at $C = 0.25$ (0.4935/0.4544 vs. 0.4351/0.3762), and the development sweep selects $C = 0.125$ (0.5195/0.4841). The reported Colab run is Table 2: $F = 0.2021$, $A = 0.5000$, $H = 0.2879$.

Context format	C	Acc.	Macro-F1
Plain	0.25	0.4351	0.3762
Rank-aware	0.25	0.4351	0.3749
Top-weighted	0.25	0.4416	0.3938
Rank-weighted	0.25	0.4351	0.3882
Enhanced	0.25	0.4935	0.4544
Enhanced	0.125	0.5195	0.4841

C	Acc.	Macro-F1
0.125	0.5195	0.4841
0.25	0.4935	0.4544
0.5	0.4740	0.4263
1.0	0.4675	0.4184
2.0	0.4416	0.3825
4.0	0.4545	0.3913
8.0	0.4481	0.3925

Table 3: Classifier ablations. Rank-aware, top-weighted, factual-cue-enhanced context improves over plain concatenation; the enhanced classifier peaks at $C = 0.125$ in the development sweep.

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